TalentFlow: Multi-Source Candidate Data Transformer

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1. Pipeline Architecture

The system is a deterministic, modular 6-stage DAG to ensure traceability and strict error isolation:

- **Ingest (Detect):** Discovers files in deterministic alphabetical order, validates MIME types, and routes to appropriate parsers.
- **Extract:** Source-specific parsers extract intermediate data. The ResumeParser uses strict regex and structural cues, prioritizing "honestly-empty" over "wrong-but-confident" by extracting only from designated sections.
- **Normalize:** Cleans extracted data (e.g., standardizing phones, mapping countries, resolving skill aliases).
- **Merge:** Resolves identity and merges data across sources using a Union-Find algorithm.
- **Confidence Score:** Evaluates data completeness, source reliability, and cross-source agreement.
- **Project-to-Output (Validate):** Reshapes the canonical internal record into a user-defined JSON schema.

2. Canonical Schema & Normalizations

The internal canonical profile provides a strict structure:

- **candidate_id** (UUID hash), **full_name** (String), **emails** (String Array), **links** (Object).
- **phones** (String Array): Normalized via the phonenumbers library to strict E.164 format. Invalid sequences drop out.
- **location** (Object): {city, region, country}. Normalized by mapping country strings ("UK") to ISO-3166 alpha-2 ("GB").
- **experience, education:** Dates are Normalized into strict YYYY-MM format using fuzzy parsing.
- **skills** (Array): Normalized against a canonical alias dictionary (e.g., "js" -> "JavaScript").
- **provenance** (Array): Tracks lineage {field, source, method} for every canonical data point.

3. Merge / Conflict-Resolution Policy & Confidence

Identity Resolution & Conflict Policy: Identity is resolved using a Union-Find graph primarily via emails (and exact name match as fallback). When merging, sources are ranked by structural trustworthiness (JSON [0.9] > CSV [0.7] > PDF [0.65] > GitHub [0.4]). If sources provide conflicting scalar values (e.g., headline), the highest-weight source strictly wins. Array fields (like skills) are unioned and deduplicated.

Confidence Formula: confidence = source_weight * completeness * agreement_bonus.
Completeness drops to 0.0 if a field is null/empty. An agreement bonus (1.2x) is applied if 2+ independent sources agree on the exact value. Overall confidence is the arithmetic mean of all field scores.

4. Runtime Configurable Output (Projection Layer)

TalentFlow decouples its internal state from the output shape using a dynamic JSON Projection Config:

- **Selection & Remapping:** Resolves nested paths via array indexing ("from": "emails[0]" maps to primary_email).
- **Format Toggles:** Applies per-field normalizations directly at output (e.g., "normalize": "E164") and toggles features like include_provenance.
- **On-Missing Policy:** Defines fallback behavior if required data is missing ("null" sets to null, "omit" removes the key entirely, "error" aborts pipeline).

5. Edge Cases & Scope Exclusions

Edge Cases Successfully Handled:

- **Garbage Documents (e.g., Financial Invoices):** Strict regex guardrails (requiring headers like "Technical Skills") prevent extracting random fragments. If unsure, the parser yields empty lists rather than false positives.
- **GitHub API Rate Limits:** If a 403 Rate Limit is hit, the parser implements a cascading fallback to raw HTML scraping to ensure continuous data extraction without crashing.
- **Array Access on Empty Arrays:** If a user config requests emails[0] but no emails were found, the projection layer gracefully applies the on_missing policy rather than throwing an IndexError.

Deliberately Descoped (Due to Time/Constraints):

- **Image-only PDF OCR:** Excluded heavy dependencies like Tesseract OCR. Pure image PDFs currently return an empty record.
- **LLM Extraction:** Opted for deterministic regex over generative AI to guarantee zero-latency execution, perfect determinism, and 100% explainability without network dependencies.